

Nftables uses a program called *nft* to add, create, list, delete and load rules. Make sure nftables is installed along with *conntrackd* and *netfilter-persistent*, and remove iptables, using the following command:

```
apt-get install nftables conntrackd
netfilter-persistent
apt-get purge iptables
```

nft needs to be run as root or use *sudo*. Use the following commands to list, flush, delete ruleset and load the script respectively.

```
nft list ruleset
nft flush ruleset
nft delete table inet filter
/usr/sbin/nft -f /etc/nftables.conf
```

Input policy

The firewall will contain three parts – input, forward and output – just like in iptables. In the terminal, type the following commands for the input firewall. Make sure you have flushed your rule-set before you begin. Our default policy will be to drop everything. We will use the inet family in the firewall. Add the following rules as root or use *sudo*:

```
nft add table inet filter
nft add chain inet filter input { type
filter hook input priority 0 \; counter
\; policy drop \; }
```

You have noticed there is something called *priority 0*. It means giving the rule higher precedence. Hooks typically give higher precedence to the negative integer. Every hook has its own precedence and the filter chain has priority 0. You can check the nftables wiki page to see the priority of each hook.

To know the network interfaces in your computer, run the following command:

```
ip link show
```

It will show the installed network interface, one local host and other Ethernet port or your wireless port. Your

Ethernet port's name looks something like this: *enpXsY* where X and Y are numbers, and the same goes for your wireless port. We have to allow the local host and only allow established incoming connections from the Internet.

Nftables has a feature called verdict statements on how to parse a rule. The verdict statements are *accept*, *drop*, *queue*, *jump*, *goto*, *continue* and *return*. Since the firewall is a simple one, we will use either *accept* or *drop the packets* (Figure 2).

```
nft add rule inet filter input iifname
lo accept
nft add rule inet filter input iifname
enpXsY ct state new, established,
related accept
```

Next, we have to add rules to protect us from stealth scans. Not all stealth scans are malicious but most of them are. We have to protect the network from such scans. The first set lists the TCP flags to be tested. Of these flags, the second set lists the flags to be

matched with the first.

```
nft add rule inet filter input iifname
enpXsY tcp flags \& \(\syn\|fin\) == \
(\syn\|fin\) drop
nft add rule inet filter input iifname
enpXsY tcp flags \& \(\syn\|rst\) == \
(\syn\|rst\) drop
nft add rule inet filter input iifname
enpXsY tcp flags \& \(\fin\|rst\) == \
(\fin\|rst\) drop
nft add rule inet filter input iifname
enpXsY tcp flags \& \(\ack\|fin\) == fin
drop
nft add rule inet filter input iifname
enpXsY tcp flags \& \(\ack\|psh\) == psh
drop
nft add rule inet filter input iifname
enpXsY tcp flags \& \(\ack\|urg\) == urg
drop
```

Remember, we are typing these commands in the terminal. So we have to add a backslash before some special characters, to make sure the terminal interprets it as it should. If you are using a script, then this isn't required.

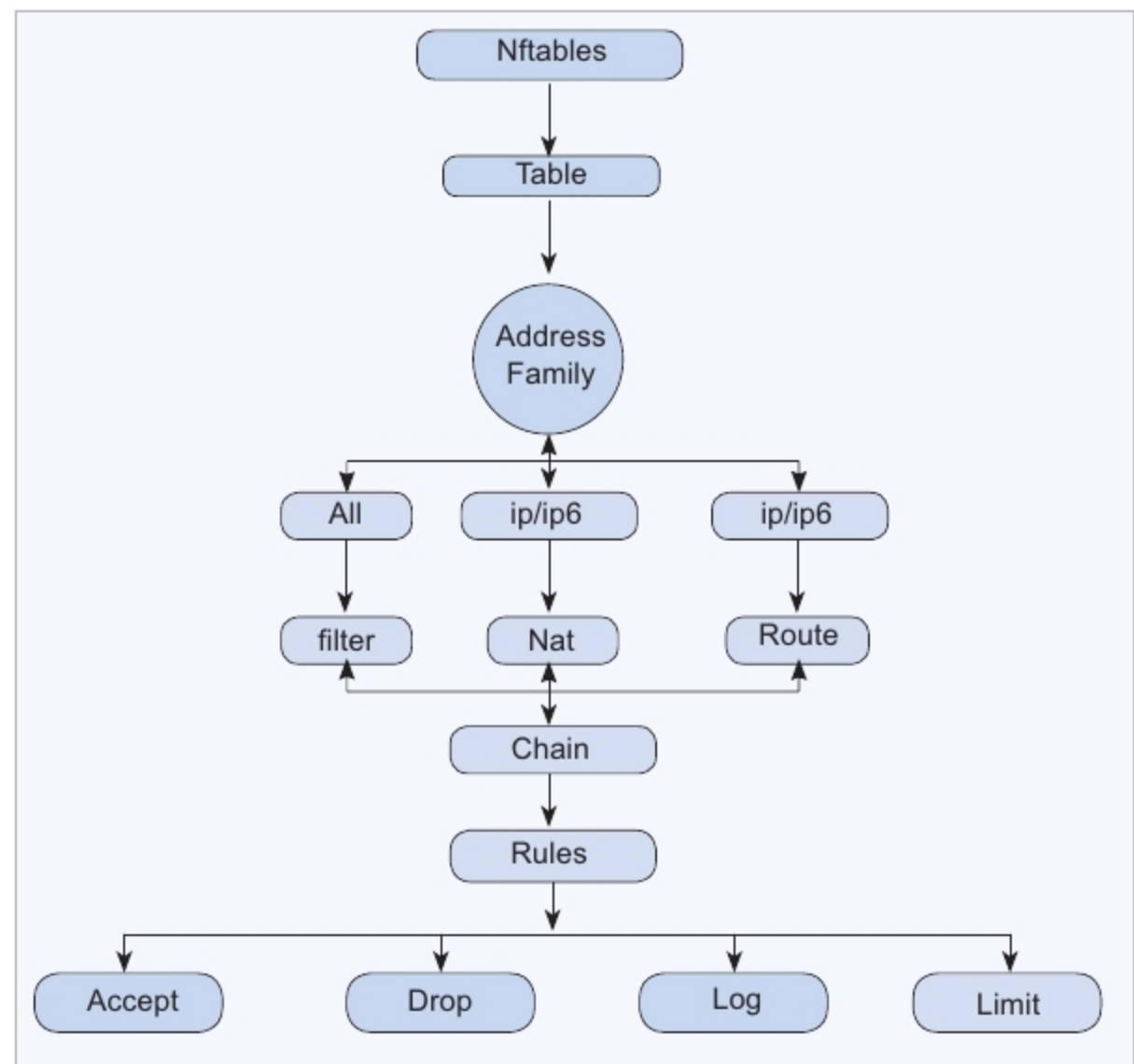


Figure 2: Simple nftables firewall